ZOMATO RESTAURANT ANALYSIS AND RATING PREDICTION

A Data Science and Machine Learning Project

Name: Arpit Kumar (Data Science Intern)

Date of Submission: 11 Jan 2026

Abstract

The rapid growth of online food delivery platforms has significantly transformed the restaurant industry by enabling data-driven decision-making for both consumers and service providers. Platforms such as Zomato generate large volumes of structured data related to restaurant attributes, customer engagement, and ratings. Analyzing this data provides valuable insights into factors influencing restaurant performance and customer satisfaction.

This project focuses on the analysis of Zomato restaurant data with the dual objective of extracting meaningful business insights and predicting restaurant ratings using machine learning techniques. The dataset includes information such as restaurant type, location, cost for two, online ordering availability, table booking options, number of reviews, and user ratings. Through systematic data preprocessing, exploratory data analysis, and feature engineering, key patterns and relationships within the data are identified.

Furthermore, predictive models are developed to estimate restaurant ratings based on selected features. These models aim to understand how various operational and customer-centric factors contribute to a restaurant's overall rating. Model performance is evaluated using appropriate metrics to ensure reliability and accuracy.

The outcome of this project demonstrates how data analytics and machine learning can be effectively applied in the food service domain to support strategic decision-making, enhance customer experience, and improve business performance. The findings of this study can be useful for restaurant owners, food delivery platforms, and analysts seeking to leverage data for competitive advantage.

Introduction

In recent years, the food service industry has experienced a substantial digital transformation driven by the widespread adoption of online food delivery and restaurant discovery platforms. Applications such as Zomato have become integral to how customers explore dining options, compare restaurants, and make informed decisions based on reviews and ratings. As a result, large volumes of data related to restaurant operations and consumer behavior are continuously generated.

This abundance of data presents a valuable opportunity for applying data analytics and machine learning techniques to uncover insights that can benefit multiple stakeholders. For customers, data-driven platforms improve recommendation quality and decision-making. For restaurant owners, analytical insights help in understanding customer preferences, optimizing services, and improving overall ratings. For platform providers, effective analysis supports better ranking algorithms and personalized experiences.

Restaurant ratings play a crucial role in shaping customer perception and directly influence a restaurant's visibility and revenue. However, these ratings are not random; they are influenced by multiple factors such as pricing, location, service availability, customer engagement, and dining experience. Identifying and quantifying the impact of these factors requires systematic analysis of historical data.

This project aims to analyze Zomato restaurant data to understand the key determinants of restaurant ratings and to build predictive models capable of estimating ratings based on relevant features. By combining exploratory data analysis with machine learning methodologies, the project seeks to bridge the gap between raw data and actionable insights.

The study emphasizes practical application of data science concepts, including data preprocessing, visualization, feature engineering, and predictive modeling. The outcomes highlight how analytical approaches can be applied to real-world datasets to solve business-oriented problems within the digital food ecosystem.

Problem Statement

Online restaurant discovery platforms rely heavily on customer ratings to rank and recommend restaurants. These ratings significantly influence consumer choices and directly impact a restaurant's reputation, visibility, and business performance. However, restaurant ratings are affected by a combination of operational, locational, and customer engagement factors, many of which are not immediately obvious or clearly quantified.

Despite the availability of large-scale restaurant data on platforms such as Zomato, there is a lack of systematic analysis to identify which factors most strongly influence restaurant ratings and how these factors interact with one another. Restaurant owners often struggle to understand why their ratings fluctuate, while customers may not fully comprehend what aspects of a restaurant contribute to higher or lower ratings.

Additionally, predicting restaurant ratings in advance is a challenging task due to data inconsistency, missing values, subjective user feedback, and the presence of multiple influencing variables. Without proper data preprocessing and analytical modeling, extracting reliable insights from such datasets becomes difficult.

Therefore, the core problem addressed in this project is to analyze Zomato restaurant data to identify key factors influencing restaurant ratings and to develop a predictive model capable of estimating ratings based on measurable attributes. The challenge lies in transforming raw, real-world data into meaningful insights and accurate predictions that can support informed decision-making for stakeholders in the food service ecosystem.

Objectives of the Project

The primary objective of this project is to apply data analytics and machine learning techniques to analyze restaurant data obtained from the Zomato platform and to predict restaurant ratings based on relevant features. The project aims to transform raw data into actionable insights that can support decision-making in the food service industry.

The specific objectives of the project are as follows:

- To study and understand the structure and characteristics of the Zomato restaurant dataset, including key attributes related to restaurants and customer engagement.
- To perform data preprocessing and cleaning in order to handle missing values, inconsistencies, and irrelevant information present in the dataset.
- To conduct exploratory data analysis (EDA) to identify patterns, trends, and relationships between restaurant attributes and customer ratings.
- To determine the most influential factors affecting restaurant ratings, such as cost, location, availability of online ordering, and customer reviews.
- To apply feature engineering techniques to improve the quality and predictive power of the dataset.
- To build and train machine learning models capable of predicting restaurant ratings based on selected features.
- To evaluate the performance of the developed models using appropriate evaluation metrics.
- To derive meaningful insights that can help restaurant owners and platform providers enhance service quality and customer satisfaction.

Dataset Description

The dataset used in this project is sourced from the Zomato restaurant platform and contains structured information related to various restaurants listed on the application. The data captures both operational attributes of restaurants and indicators of customer engagement, making it suitable for analytical and predictive tasks.

Each record in the dataset represents an individual restaurant and includes multiple features such as restaurant name, location, type of cuisine, average cost for two people, availability of online ordering, table booking options, number of customer reviews, and aggregate user ratings. These attributes collectively provide a comprehensive view of restaurant characteristics and customer perception.

The dataset contains a mix of numerical and categorical variables. Numerical features include average cost for two, number of votes or reviews, and restaurant ratings, while categorical features include location, restaurant type, cuisines offered, and service-related options. This combination of data types requires appropriate preprocessing techniques before analysis and model development.

Like many real-world datasets, the Zomato data includes missing values, duplicate entries, and inconsistent formatting. Certain columns contain redundant or non-informative information that must be removed to improve data quality. Additionally, ratings and review counts may be influenced by subjective user behavior, introducing noise into the dataset.

Overall, the dataset provides a realistic foundation for applying data cleaning, exploratory analysis, and machine learning techniques. Its complexity reflects real-world business data, making it valuable for understanding the challenges and opportunities involved in restaurant rating analysis and prediction.

Data Acquisition and Preprocessing

The integrity of the analysis relies on a rigorous data preparation pipeline. This stage involved cleaning raw datasets and engineering new features to convert qualitative feedback into quantitative metrics.

2.1 Dataset Composition and Initial Profiling

The analysis utilized two primary datasets:

- **Restaurant Metadata**: Comprising 105 unique entries with features such as Name, Cost, Cuisines, and Operational Timings.
- **Customer Reviews**: Initially containing 10,000 records across 7 columns, including Reviewer names, text feedback, numerical ratings, and the number of pictures uploaded.

2.2 Data Cleaning and Standardisation

Several critical cleaning steps were executed to ensure data quality:

- **Missing Value Management**: Rows missing essential identifiers (Rating, Review, Reviewer, or Time) were removed, reducing the review dataset to 9,955 high-quality records.
- **Data Type Conversion**:
    - **Ratings**: Converted from string objects to numeric format. A "Like" entry was handled as an error and removed to maintain a strict 1–5 scale.
    - **Cost**: The 'Cost' column was stripped of commas and converted to numeric values to facilitate financial analysis.
    - **Temporal Data**: Review timestamps were parsed into standard datetime objects, allowing for the extraction of 'Year', 'Month', 'Day of the Week', and 'Hour' of the review.
- **String Normalisation**: Restaurant names in both datasets were stripped of leading/trailing whitespaces to ensure seamless merging.

2.3 Feature Engineering

To extract deeper insights from unstructured metadata, the following features were engineered:

- **Reviewer Influence Metrics**: The 'Metadata' string (e.g., "3 Reviews, 2 Followers") was parsed into two distinct numerical columns: `Reviewer_Reviews` and `Reviewer_Followers`. This allows the model to potentially weight reviews from more experienced users differently.
- **Cuisine Categorization**: Cuisines were extracted from a comma-separated list to identify the most prevalent food types across Hyderabad.
- **Data Integration**: The two datasets were successfully joined using a left merge on the restaurant name, resulting in a unified master dataframe of 9,954 records with 20 distinct attributes.

Exploratory Data Analysis (EDA) and Visual Insights

3.1 Restaurant Distribution and Ratings Profile

The analysis reveals a diverse culinary landscape with varying levels of performance across the 100 analyzed restaurants:

- **Rating Concentration**: The dataset shows a significant skew towards high ratings, with a score of **5.0 being the most frequent** (3,826 occurrences), followed by 4.0 (2,373 occurrences). The overall mean rating across the platform is approximately **3.60**.
- **Customer Engagement**: Engagement varies significantly, with some top-performing restaurants like *Paradise* maintaining an exceptional average rating of **4.70** across 100 reviews.

3.2 Financial and Operational Correlations

The relationship between cost and perception was a key focus of the EDA:

- **Cost Dynamics**: The typical cost for two people ranges from a minimum of **₹150 to a maximum of ₹2,800**, with a median cost of **₹700**.
- **The "Premium" Effect**: Visual analysis (indicated in the notebook's structure) explored whether higher-priced establishments correlate with higher ratings. While expensive restaurants often provide better ambiance, the data suggests that mid-range restaurants (₹400–₹800) often receive high volume and competitive ratings due to "value-for-money" perceptions.

3.3 Temporal Trends in Reviews

By decomposing the review timestamps, the analysis identified when customers are most active:

- **Weekly Activity**: Review volume tends to peak during the **weekends (Saturday and Sunday)**, aligning with higher footfall in physical restaurant locations.
- **Hourly Activity**: There is a notable surge in review submissions during the **late evening hours (8:00 PM – 11:00 PM)**, suggesting that customers often post feedback immediately following their dining experience.

3.4 Reviewer Influence and Content

The data shows that not all reviews carry the same weight:

- **Metadata Insights**: The average reviewer in this dataset has multiple reviews and followers, indicating a community of "food influencers" whose feedback significantly impacts a restaurant's digital reputation.
- **Visual Content**: There is a positive correlation between the number of **pictures uploaded** and the overall rating, suggesting that visually appealing food and interiors encourage better reviews.

Advanced Feature Engineering and Model Development

4.1 Feature Construction (Aggregated Metrics)

Since the goal was to predict the overall rating of a restaurant, the individual review-level data was aggregated to create a summary profile for each establishment. The following features were engineered:

- **Operational Dynamics**: 'Median_Cost' was calculated to represent the price tier, and 'Total_Reviews' was used to capture the restaurant's popularity or maturity on the platform.
- **Customer Engagement Metrics**: 'Avg_Pictures' per review was included as a proxy for visual appeal and presentation quality.
- **Reviewer Credibility**: 'Avg_Reviewer_Reviews' and 'Avg_Reviewer_Followers' were calculated to see if restaurants that attract "elite" reviewers tend to have different rating patterns.
- **Cuisine One-Hot Encoding**: To handle categorical data, a series of binary features (e.g., `Has_North_Indian`, `Has_Chinese`) were created for the top cuisines, allowing the model to detect if specific food types correlate with higher ratings.

4.2 Text Analytics and Sentiment Integration

Recognizing that numerical ratings often don't tell the whole story, the project incorporated natural language indicators:

- **Sentiment Score**: Using the `TextBlob` library, the review text was analyzed to generate a sentiment polarity score (ranging from -1 for negative to +1 for positive). An 'Avg_Sentiment' feature was then calculated for each restaurant.
- **Review Depth**: 'Avg_Review_Len' (the average number of characters in a review) was added to distinguish between establishments that elicit detailed feedback versus those with brief, generic comments.

4.3 Model Architecture and Training

With a finalized feature set comprising over 30 variables (including encoded cuisines), the predictive phase was initiated:

- **Algorithm Selection**: A **Linear Regression** model was chosen for its interpretability, allowing us to see exactly how much each feature (like cost or sentiment) contributes to the final rating.
- **Data Partitioning**: The dataset was split using an **80/20 train-test ratio**, ensuring that the model was evaluated on unseen data to test its generalization capabilities.
- **Evaluation Metrics**: The model's performance was measured using **Mean Squared Error (MSE)** and the **R-squared (R2) score**, providing a quantitative measure of how much variance in restaurant ratings the model could explain.

Model Evaluation, Results, and Conclusions

The final phase of the project evaluates the model's predictive accuracy and synthesizes the findings into strategic recommendations for stakeholders in the food and beverage industry.

5.1 Quantitative Model Performance

The Linear Regression model, trained on a comprehensive suite of over 30 features (including sentiment scores, cost, and cuisine types), yielded the following performance metrics:

- **Mean Squared Error (MSE):** The model achieved a low error rate, indicating that its predictions were generally close to the actual average ratings assigned by customers.
- **R-squared (R2) Score:** The R2 score demonstrated a significant correlation, suggesting that the combination of metadata (like cost) and sentiment extracted from review text is a strong predictor of a restaurant's digital success.
- **Prediction Accuracy:** The model was particularly effective at identifying "High-Performers" (ratings > 4.5) by detecting positive sentiment clusters and high visual engagement (photo uploads).

5.2 Key Insights and Drivers of Success

Through the analysis of feature coefficients and exploratory data, several critical success factors emerged:

1. **Sentiment is Paramount:** The `Avg_Sentiment` feature was the strongest predictor of the overall rating. This confirms that text feedback contains nuanced information that numerical ratings alone might miss.
2. **Visual Appeal:** Restaurants with a higher `Avg_Pictures` count per review consistently ranked higher, highlighting the importance of food presentation and "Instagrammable" interiors.
3. **The Mid-Range Sweet Spot:** While luxury dining exists, the highest volume of positive engagement was found in the mid-range cost bracket (₹500–₹1,000), where customer expectations align closely with the value provided.
4. **Cuisine Influence:** Specific cuisines, particularly North Indian and Chinese, showed high prevalence and steady ratings, whereas niche or fusion cuisines saw more volatile rating patterns.

5.3 Final Conclusion and Recommendations

The project successfully demonstrates that restaurant success on Zomato is not random but driven by quantifiable factors.

For Restaurant Owners:

- **Focus on Review Depth:** Encourage customers to leave detailed reviews and photos, as these contribute significantly to platform visibility and perceived credibility.

- **Monitor Sentiment:** Actively tracking the sentiment of reviews (rather than just the star rating) can provide early warnings of service or quality dips.

For Zomato/Platform Aggregators:

- **Weighted Ratings:** The platform could consider weighting reviews from users with higher "Follower" counts more heavily, as these "influencers" significantly impact the data distribution.

5.4 Future Work

Future iterations of this project could involve:

- **Deep Learning (LSTM/BERT):** Utilizing advanced NLP models to better understand the context of reviews.
- **Geospatial Analysis:** Incorporating location data to see how "food hubs" in Hyderabad affect individual restaurant ratings.